

STATION 1 SCANNER APP

Operating procedure for the PPNAM Station 1 handheld RFID scanner — operator login, RFID tag assignment, pallet offloading against SAP documents, and device configuration.

SYSTEM PPNAM Station 1 AA (Android)	DEVICE Chainway C72 RFID handheld	APP PACKAGE com.mitas.ppnam.station1aa
APP VERSION v1.0 (1)		

STEP 1
LOG IN

STEP 2
DASHBOARD

WORKFLOW A
TAG ASSIGNMENT

WORKFLOW B
OFFLOAD

1 OVERVIEW

The Station 1 Scanner App runs on the Chainway C72 handheld and gives goods-receiving operators two workflows against the **Station 1 PC**:

- **Tag Assignment** — register RFID pallet tags with the station. Scanning a tag sends it automatically; the station accepts or rejects each tag.
- **Offload** — record each pallet coming off the truck against an SAP document (Purchase Order or Stock Transfer): scan the pallet's RFID tag and barcode, confirm the pallet values, and close the document when the delivery is complete.

Every operator signs in first — with a username and password, or by scanning their badge. What an operator can open is controlled per account: a tile the operator is not authorised for stays greyed out on the dashboard.

The scanner talks to the Station 1 PC over the site Wi-Fi using MQTT. The station validates every scan and every login. If the station goes offline, the app blocks all work with a full-screen overlay until it reconnects ([Section 8](#)).

2 PREREQUISITES

Confirm all of the following before starting a shift:

- ✓ The scanner is charged and powered on.
- ✓ Wi-Fi is connected to the site network.
- ✓ The Station 1 PC is running and online.
- ✓ The app is installed: `com.mitas.ppnam.station1aa`.
- ✓ The handheld has been provisioned with its broker credential (one-time supervisor task — see [Section 7](#)). The connection pill in the top bar reads **Connected**.
- ✓ The operator has a user account or badge registered at Station 1.

3 LOGGING IN

STEP 1

The app opens on the Log In screen. Check the connection pill at the top right reads **Connected** before logging in.

There are two ways to sign in:

USERNAME + PASSWORD

Type your operator username and password, then tap **LOG IN**. Credentials are verified by the Station 1 PC (SCRAM challenge — the password itself is never sent over the network).

BADGE SCAN

Simply scan your badge with the RFID trigger while on the Log In screen — no typing needed.

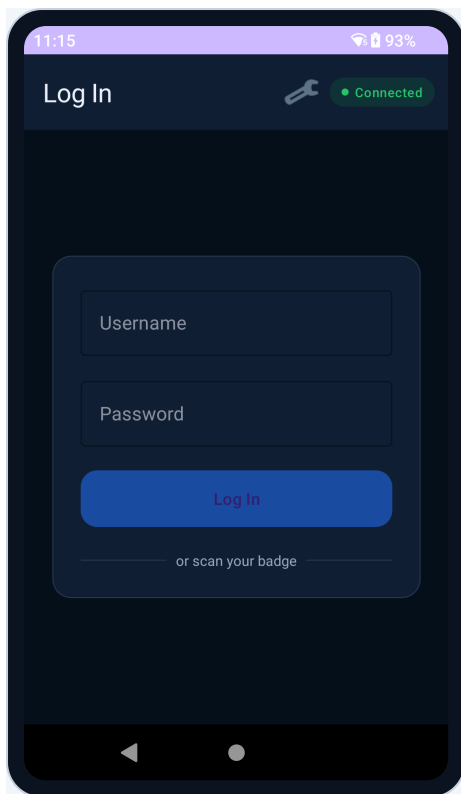


FIG 3.1
Log In screen. Pill shows Connected; the ⚙ icon opens Settings.

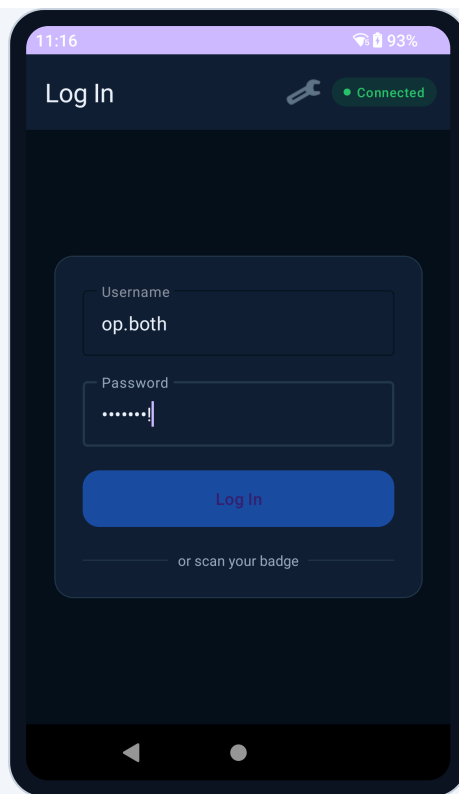


FIG 3.2
Credentials entered — tap LOG IN.

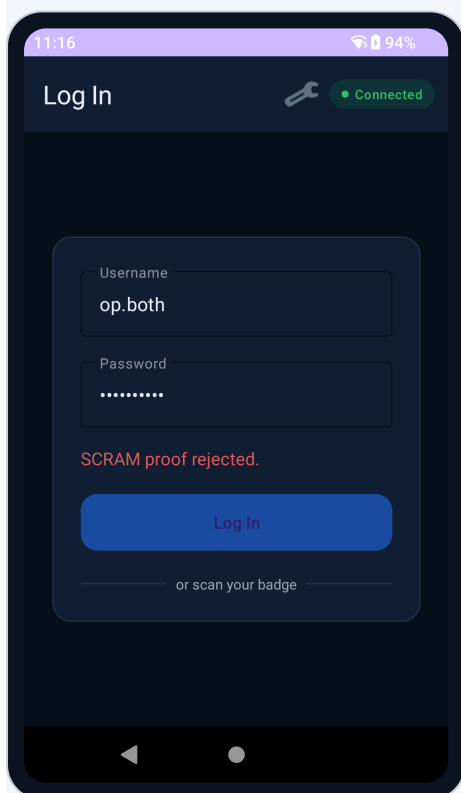


FIG 3.3
Rejected login — a red error (e.g. "SCRAM proof rejected.") appears and you stay on this screen.

3.1 IF THE LOGIN FAILS

- **Wrong password or unknown username** — a red error appears under the password field. Re-type the credentials and try again.
- **Unknown badge** — the same red error appears. Verify the badge is registered at Station 1.
- **No response from the station** — the request times out after ~10 seconds and shows an error; the screen stays usable. Try again — if it keeps happening, check [Section 8](#).

4 DASHBOARD & OPERATORS

STEP 2

After login the dashboard shows who is signed in and the two workflow tiles. Tap a tile to open that workflow; tap the **back arrow** inside a workflow to return here.

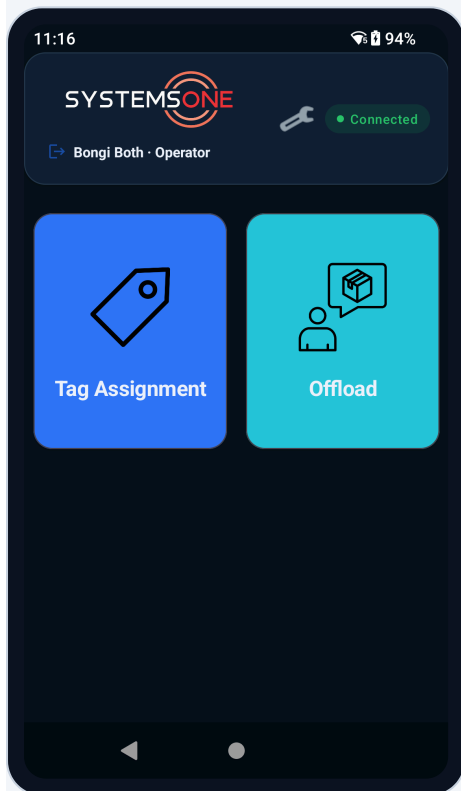


FIG 4.1
Dashboard — operator "Bongi Both"
with both tiles enabled.

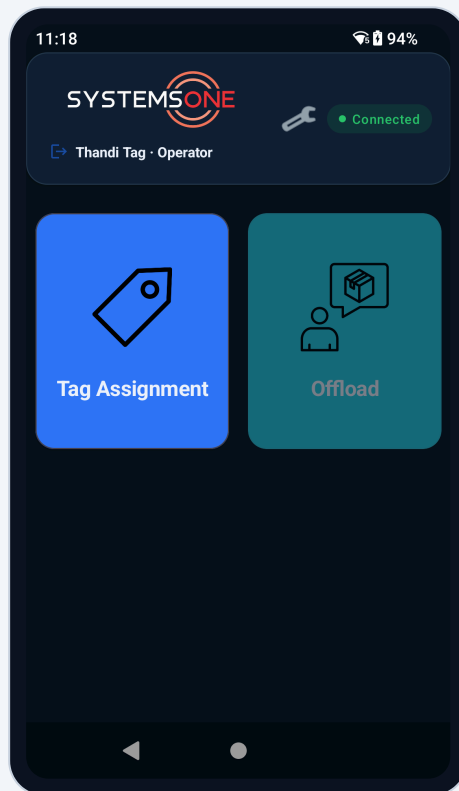


FIG 4.2
Permissions are per operator — this
account has Tag Assignment only, so
Offload is greyed out.

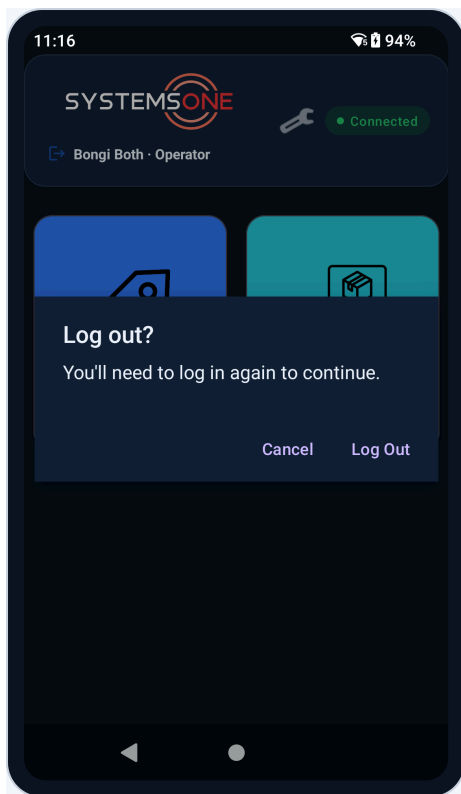


FIG 4.3

Tapping the operator name prompts to log out.

OPERATOR ROW	Shows the signed-in operator. Tap it to log out (confirm in the dialog). Settings offers a second Log Out button.
TAG ASSIGNMENT TILE	Opens Workflow A (Section 5). Enabled only if your account allows it.
OFFLOAD TILE	Opens Workflow B (Section 6). Enabled only if your account allows it.
⚙️ / PILL	The gear opens Settings; the pill shows the live broker connection (see Section 8).

NOTE Always log out at the end of your shift or when handing the scanner to another operator — every scan is recorded against the signed-in operator. If the station expires your session, the app returns to the Log In screen automatically; just log in again.

5 TAG ASSIGNMENT

Use Tag Assignment to register RFID pallet tags with Station 1. There is nothing to type — scanning is the whole workflow.

- 1 On the dashboard, tap **Tag Assignment**.
- 2 Point the scanner at the pallet tag and pull the trigger. The tag is **sent automatically** — no submit button.
- 3 Watch the **Last Scanned** card: the tag ID appears with a status underneath. Green means assigned; red means rejected (see 5.1).
- 4 Repeat for every tag. The card always shows the most recent scan.

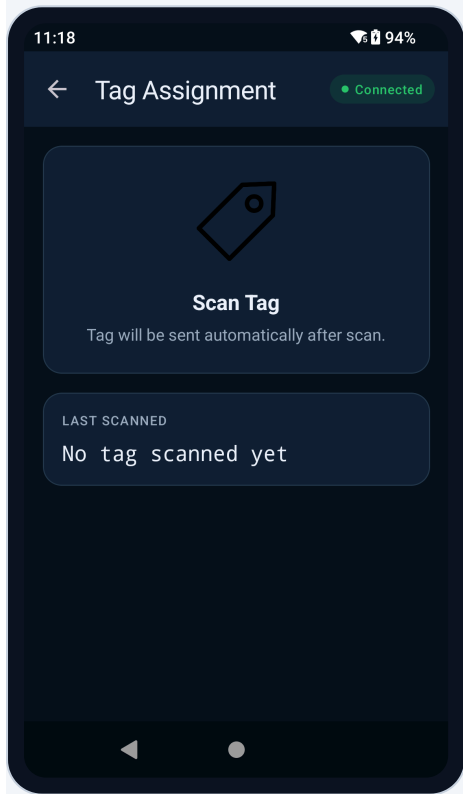


FIG 5.1
Ready to scan — "Tag will be sent automatically after scan."

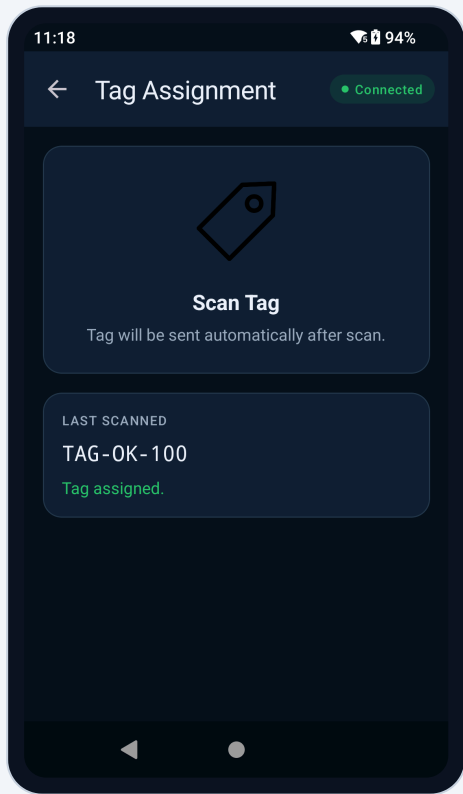


FIG 5.2
Success — **Tag assigned.** shows under the tag ID.

5.1 REJECTED OR FAILED SCANS

A red status under the tag means the station did not accept it:

TAG ALREADY IN USE.	Another scanner holds this tag. Use a different tag, or have the other operator release it.
TAG UNKNOWN.	The tag is not in the station's tag pool. Set it aside and report it to your supervisor.
NO RESPONSE FROM STATION — TRY AGAIN	The station did not answer within 10 seconds. Scan the same tag again — a successful retry assigns it normally.

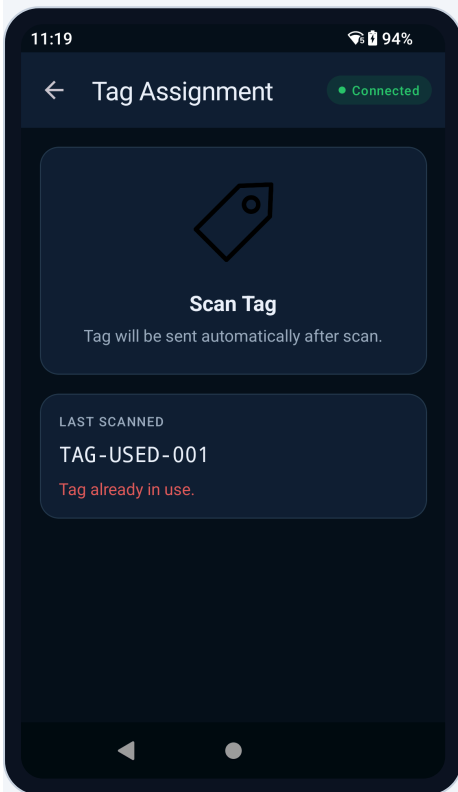


FIG 5.3
Rejected — the tag is already held by another scanner.

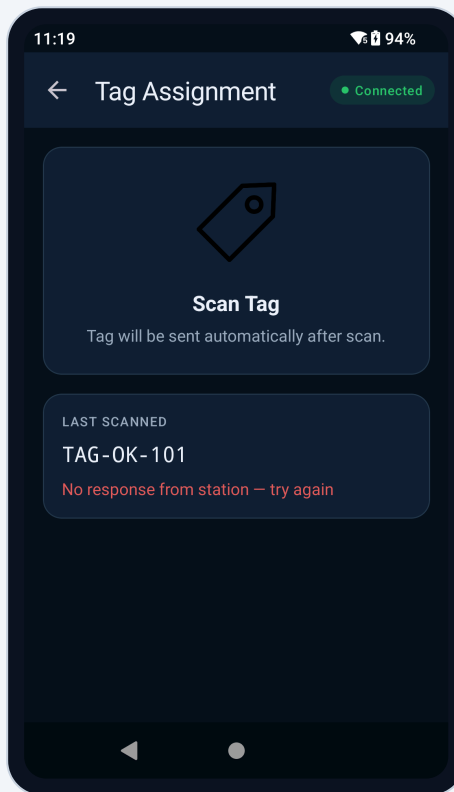


FIG 5.4
Timeout — scan the same tag again to retry.

TIP Re-scanning a tag you already own is harmless — the station simply confirms it as assigned again.

6 OFFLOAD

WORKFLOW B

Use Offload as pallets come off the truck. Each pallet carries an **RFID tag** and a **barcode label**; the station checks that the two belong to the same pallet and tells

you which SAP document it belongs to.

6.1 SCAN & MATCH

- 1 On the dashboard, tap **Offload**.
- 2 Scan the pallet's **RFID tag** and its **barcode** (either order — each scan fills its own field automatically; you can also type them).
- 3 Tap **MATCH PALLET**. The button enables once both fields are filled.

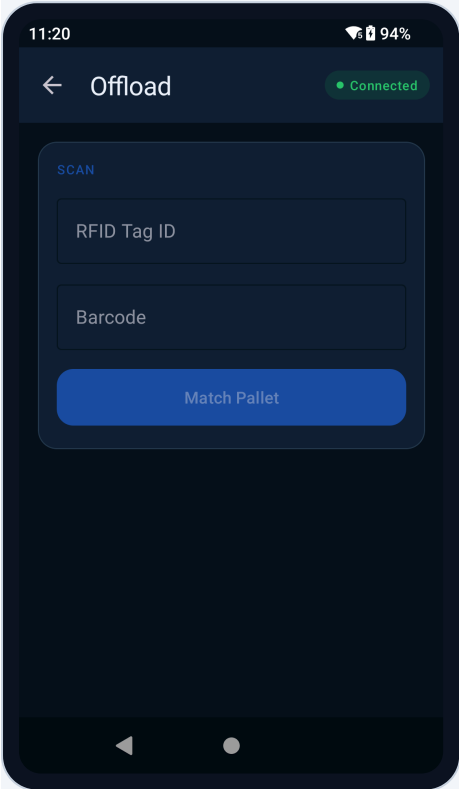


FIG 6.1
Scan step — waiting for the tag and barcode.

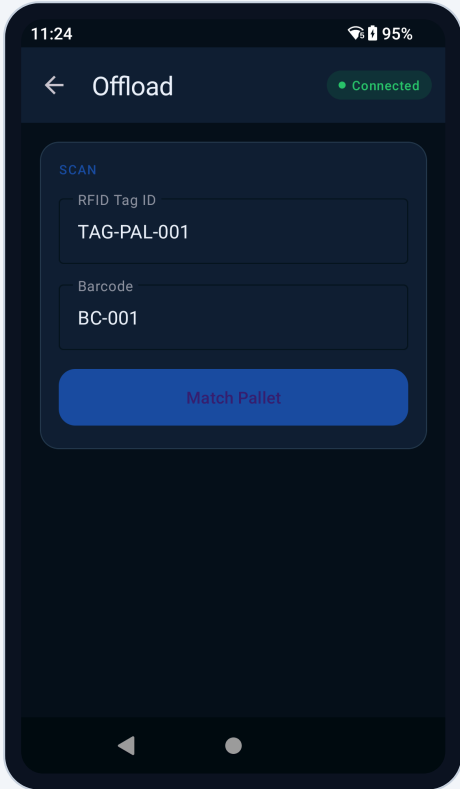


FIG 6.2
Both scanned — tap MATCH PALLET.

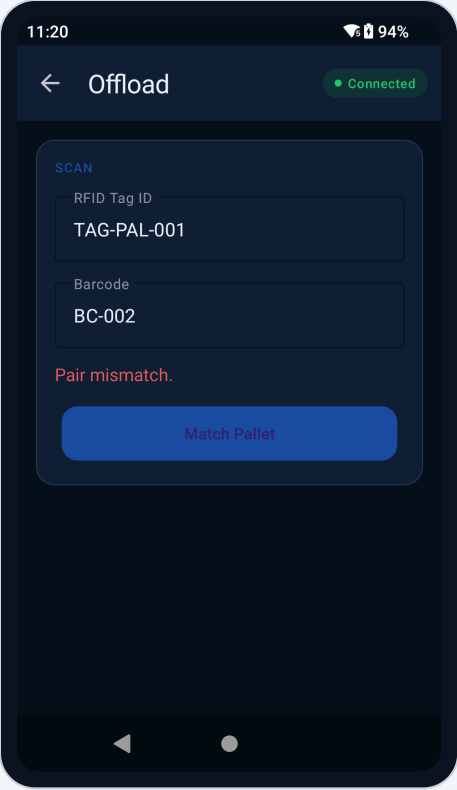


FIG 6.3
Rejected — "Pair mismatch." means the tag and barcode are from different pallets. Re-scan both on the *same* pallet.

If the station rejects the pair, the reason appears in red and you stay on the scan step:

PAIR MISMATCH.	Tag and barcode belong to different pallets — re-scan the label and tag on the same pallet.
BARCODE NOT FOUND	The label is not in the system. Check you scanned the pallet label, not another sticker.
ALREADY OFFLOADED	This pallet was recorded earlier — it does not need scanning again.
DOCUMENT UNKNOWN / CLOSED	The pallet's document is not open — see 6.4, or call your supervisor.

6.2 CONFIRM THE PALLET VALUES

On a successful match, the **Pallet Values** card appears. The heading shows the SAP document and progress (e.g. *PO-000123 · 5 of 12 pallets*), and the station pre-

fills the expected values.

- 1 Check **Bag Weight (kg)**, **Number of Bags**, and **Batch Reference** against what is physically on the pallet. Correct any of them if the pallet differs.
- 2 Tap **CONFIRM OFFLOAD**. (Use **BACK TO SCAN** to abandon this pallet — the scanned pair is kept.)

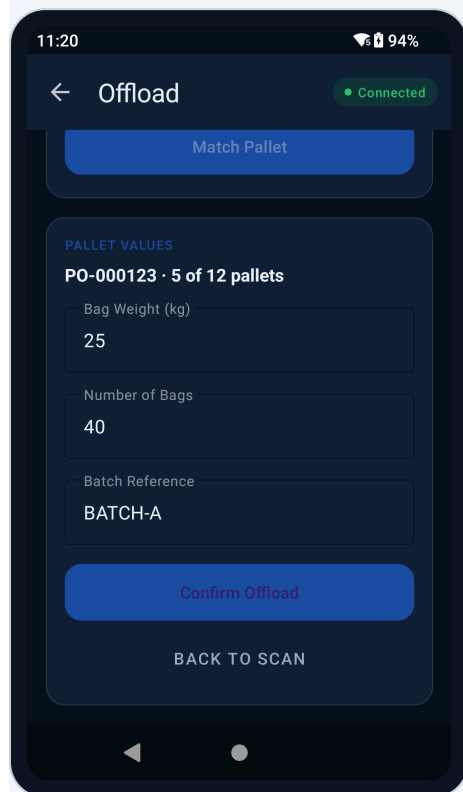


FIG 6.4

Pallet Values — PO-000123, 5 of 12 pallets, values prefilled by the station.

The app blocks obviously wrong values before sending: weight and bag count must be greater than zero and the batch reference cannot be empty. If the *station* rejects a value, the reason shows in red and you stay on this card to fix it; if there is no response, wait for the timeout message and tap CONFIRM OFFLOAD again.

6.3 PALLET RECORDED — NEXT PALLET OR DONE

On success a dialog confirms the pallet and shows the updated progress:

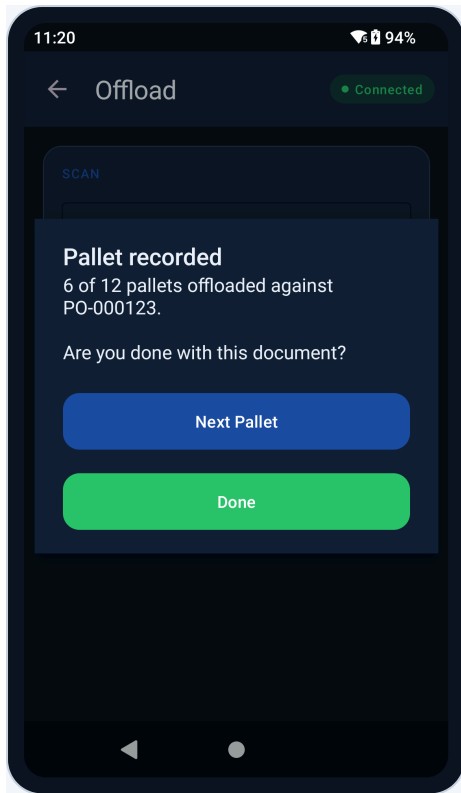


FIG 6.5

"Pallet recorded — 6 of 12 pallets offloaded against PO-000123." Blue continues, green closes.

NEXT PALLET	The everyday action — clears the scan fields for the next pallet on the truck.
DONE	Only when the delivery for this document is finished — opens the document-closing dialog (6.4).

6.4 CLOSING THE DOCUMENT

Tapping **Done** asks how to close the document. Pick the outcome that matches what was actually received — the colors mirror the meaning:

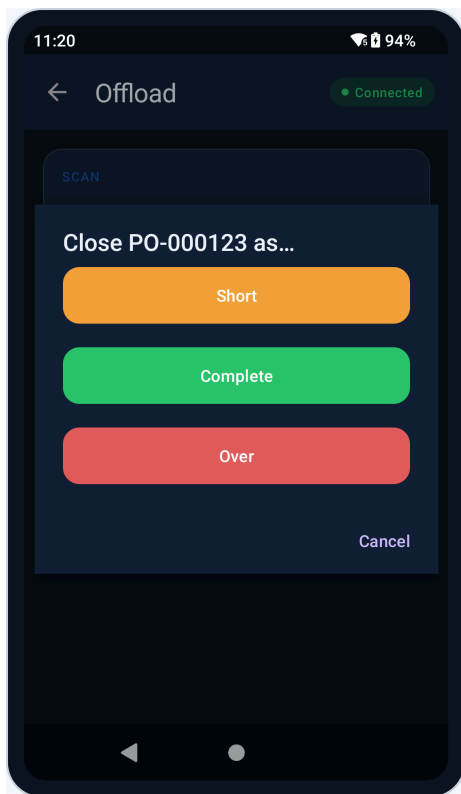


FIG 6.6

Close PO-000123 as... Short (amber), Complete (green), or Over (red).

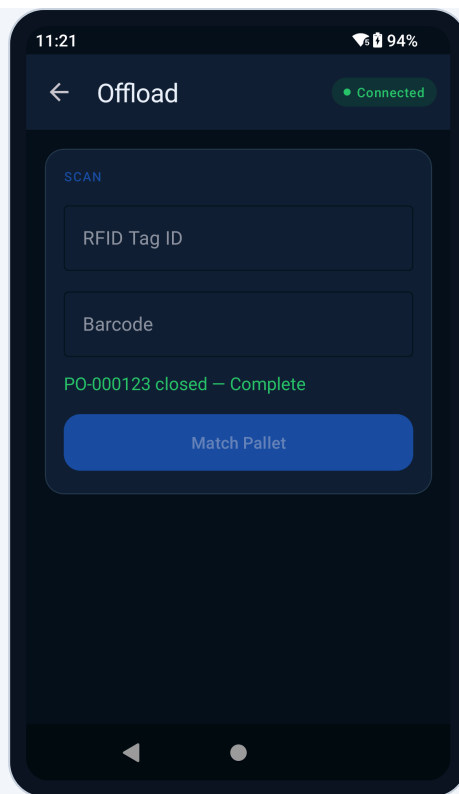


FIG 6.7

Closed — "PO-000123 closed — Complete" and the screen is ready for the next document's pallets.

SHORT	Fewer pallets arrived than the document expects.
COMPLETE	Everything expected was received.
OVER	More arrived than expected.

CAUTION Closing is final for this delivery: pallets belonging to a closed document can no longer be offloaded. If the close fails (station error), the dialog is offered again — just pick the outcome once more. **Cancel** backs out without closing.

7 SETTINGS

SUPERVISOR

Open Settings with the **gear** in the top bar (available from the Log In screen and the dashboard). Diagnostics are visible to everyone; changing the configuration requires the supervisor PIN.

7.1 DIAGNOSTICS

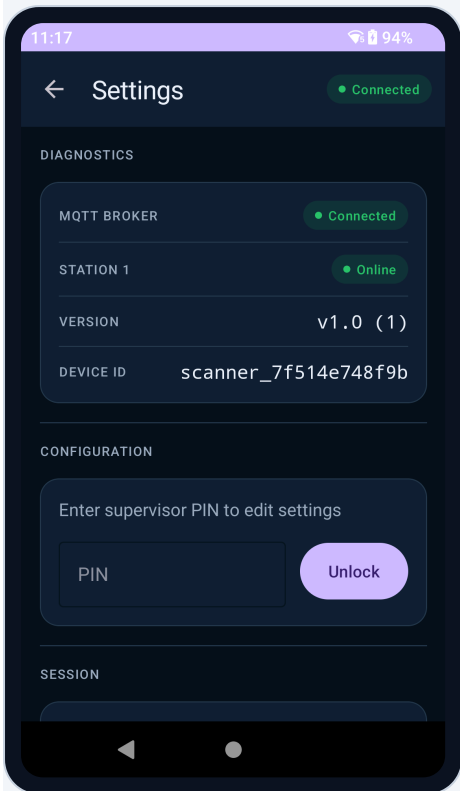


FIG 7.1
Diagnostics on top: broker Connected, Station 1 Online, app version, device ID. Below it, the PIN gate.

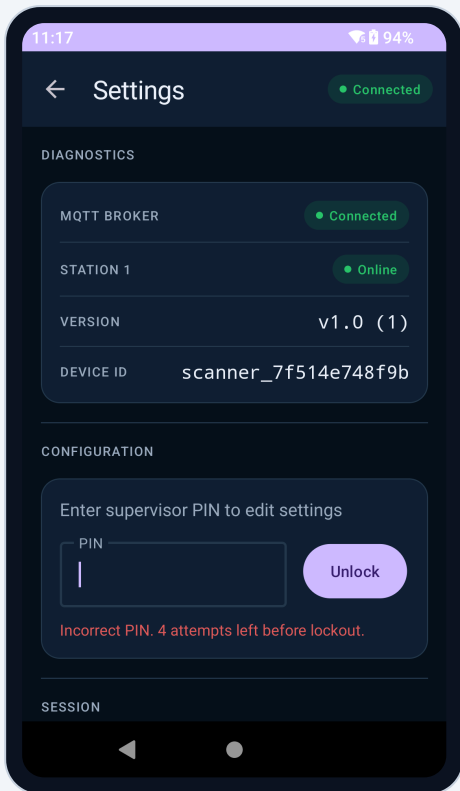


FIG 7.2
Wrong PIN — "Incorrect PIN. 4 attempts left before lockout."

MQTT BROKER	This handheld's own connection to the broker — Connected is required for everything.
STATION 1	Whether the Station 1 PC is online. Can be Offline even while the broker is Connected.
VERSION / DEVICE ID	Quote both when reporting a problem (e.g. v1.0 (1), scanner_7f514e748f9b).

LOCKOUT Five wrong PINs lock the gate for 30 seconds — the correct PIN is refused during the lockout. Wait it out, then try again.

7.2 BROKER CONFIGURATION & PROVISIONING

After unlocking with the PIN, the **Broker** form appears with the current settings. This is also how a new handheld is **provisioned**: until a broker password has been saved once, the app deliberately stays offline.

- 1 Enter the **Host**, **Port**, and the **Use WebSocket** / **Use TLS** switches as instructed by IT.
- 2 Enter the broker **Username** and **Password**. The password field is always shown empty — leaving it blank keeps the already-saved credential; typing a value replaces it.
- 3 Tap **SAVE & RESTART**. The app restarts, reconnects with the new settings, and returns to the Log In screen.

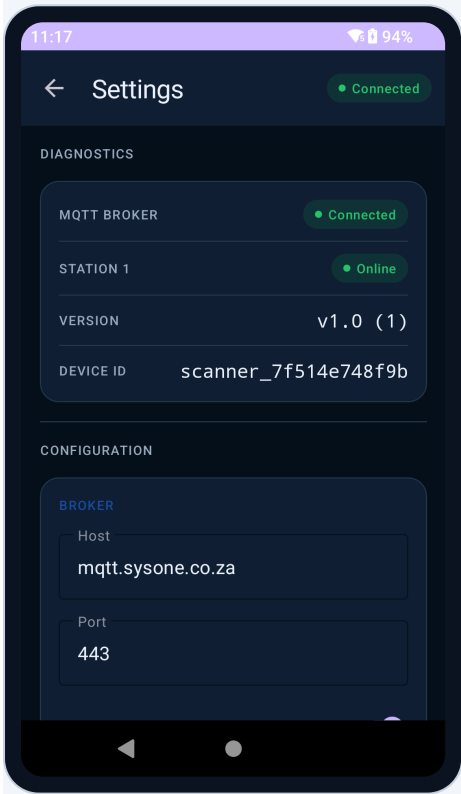


FIG 7.3
Unlocked — broker Host and Port prefilled with the current values.

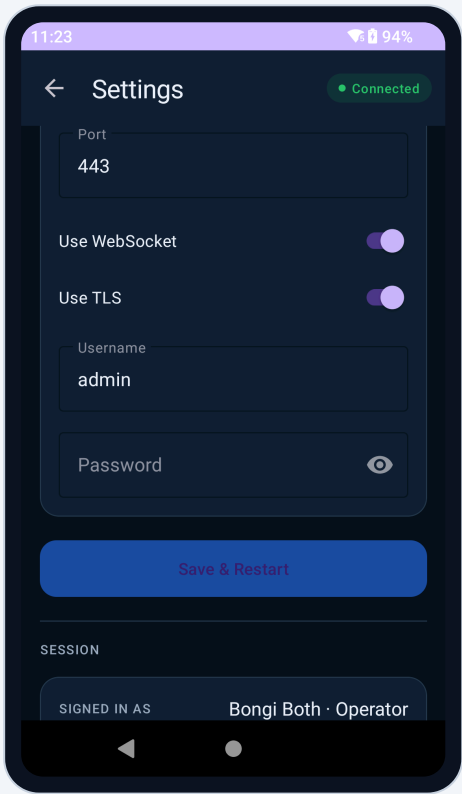


FIG 7.4
Transport switches, credentials, and SAVE & RESTART. The saved password is never displayed.

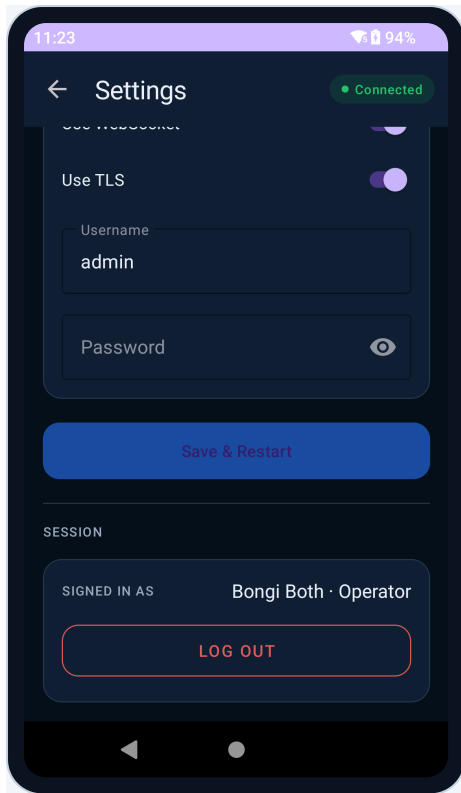


FIG 7.5
Session card — who is signed in, with a LOG OUT button.

NOTE The broker password is stored encrypted on the device. An invalid value (e.g. port 0) is refused locally — fix the field and save again. After a factory reset or app-data wipe the credential is gone and the handheld must be provisioned again.

8 CONNECTIVITY & OFFLINE

The pill in the top bar shows this scanner's broker connection on every screen:

CONNECTED

Normal — all workflows available.

CONNECTING

The app is (re)establishing the broker connection — wait a moment.

OFFLINE

No broker connection — check Wi-Fi; if it persists, check the broker settings ([Section 7](#)) or whether the device was ever provisioned.

8.1 MAIN STATION OFFLINE

If the Station 1 PC itself goes offline, the app blocks all work with a full-screen overlay (the pill reads "Station 1 offline"). Scans and logins are impossible until the station returns.

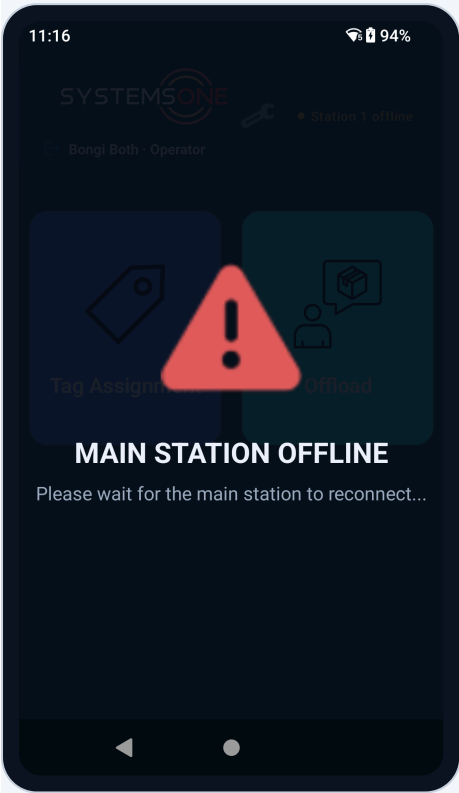


FIG 8.1
MAIN STATION OFFLINE — clears automatically when the station reconnects.

NOTE The overlay clears by itself the moment Station 1 comes back online — no action is needed on the scanner. If it does not clear, confirm the Station 1 PC is running, then check this scanner's own connection (pill and Settings diagnostics).

9 TROUBLESHOOTING

Quick reference for every message an operator may see:

WHERE	MESSAGE / SYMPTOM	MEANING	ACTION
Log In	"SCRAM proof rejected." / login error	Wrong password, unknown user, or unknown badge	Re-enter credentials; verify the account or badge (3.1)

WHERE	MESSAGE / SYMPTOM	MEANING	ACTION
Log In	Timeout error after ~10 s	Station did not answer	Try again; check Section 8
Any screen	Returned to Log In unexpectedly	Operator session expired at the station	Log in again and continue
Tag Assignment	"Tag already in use."	Another scanner holds the tag	Use another tag or have it released (5.1)
Tag Assignment	"Tag unknown."	Tag not in the station's pool	Set the tag aside; tell your supervisor
Tag / Offload	"No response from station — try again"	Request timed out	Repeat the same scan or tap the button again
Offload	"Pair mismatch."	Tag and barcode from different pallets	Re-scan both on the same pallet (6.1)
Offload	Barcode not found / already offloaded	Wrong label, or pallet already recorded	Check the label; skip an already-recorded pallet
Offload	Document unknown / closed	The pallet's document is not open	Call your supervisor — the document may be closed (6.4)
Settings	"Incorrect PIN ... before lockout"	Wrong supervisor PIN; 5 misses = 30 s lockout	Wait out the lockout, then enter the correct PIN (7.1)
Top bar	Pill shows Offline	No broker connection	Check Wi-Fi and broker settings; provision if new (7.2)
Whole screen	MAIN STATION OFFLINE overlay	Station 1 PC is down	Wait — it clears automatically when the station returns (8.1)

PPNAM-SOP-001 · Rev. B · Station 1 Scanner App

Device: Chainway C72 · Package: `com.mitas.ppnam.station1aa` · v1.0 (1)

Screenshots captured from the live app, 31 Aug 2026